Development environments, editing features and translators

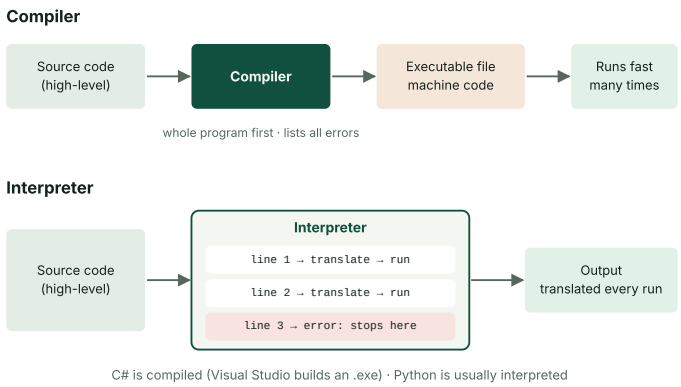
D02 FACT FILE

Specification: students should be able to

- Explain the significance of the following aspects of software development environments: editing features; high-level code translation and execution

CCEA fact file: Software Development Trends

Key words: source code, machine code, high-level language, IDE, code editor, IntelliSense, auto-complete, code outlining, line numbering, syntax highlighting, debugger, translator, compiler, interpreter



A compiler translates the whole program into an executable; an interpreter translates and runs one line at a time

Why translation is needed

High-level languages let programmers concentrate on solving the problem rather than on how the processor works. The processor, however, only executes machine code, so every high-level program must be translated. A compiler translates the whole program into an executable file before it runs; an interpreter translates and runs it one statement at a time.

Integrated development environments

An IDE combines a code editor, translator, debugger, run-time environment and often a GUI designer in a single package. Editing features such as IntelliSense, auto-complete, syntax highlighting, error underlining, line numbers and code outlining help programmers write code faster and with fewer errors. Debugging tools such as breakpoints and stepping let the programmer watch variables change, which helps find logic errors.

T Key terms from the mark schemes

IDE
Integrated development environment: editor, translator, debugger and tools in one application.
Credited: U4 2019 Q1, 2022 Q3, 2025 Q3

Translator
Software that converts source code into machine code: a compiler or interpreter.
Credited: U4 2019 Q1

Compiler
Translates the whole program into machine code before running, producing a separate executable.
Credited: U4 2022 Q3, 2024 Q5

IntelliSense / auto-complete
Editor feature that suggests or completes code as it is typed.
Credited: U4 2019 Q1, 2025 Q3

Machine code
Binary instructions the processor executes directly.
Credited: U4 2019 Q1, 2022 Q3

Interpreter
Translates and executes source code one line at a time.
Credited: U4 2024 Q5

Exam focus

You may be asked to name editor features (2019 Q1(d), 2025 Q3(c)(i)), complete a paragraph on translation (2022 Q3(a)), judge compiler/interpreter statements (2024 Q5(b)) or write a QWC answer on development environments (2023 Q8). Always say how the feature helps the programmer.

Unit 4 2025 Q3	[15]	Musician hire algorithm; variables and data types; IDE aids; syntax vs logic
Unit 4 2024 Q5	[9]	Paradigms fill-in; compilers and interpreters true/false
Unit 4 2023 Q8	[6]	Editing features, translation and execution (QWC)
Unit 4 2022 Q3	[6]	Translating code; IDE features
Unit 4 2019 Q1	[18]	Data types, range and presence checks, extending a program, IDE features, translation

? Check yourself

1. Explain why source code written in a high-level language must be translated. [2]
2. Describe two editing features of an IDE and how each helps a programmer. [4]
3. Describe how a compiler translates a program. [2]

Answers: 1. The CPU can only execute machine code (binary instructions) (1) / so high-level source code, which is close to English, must be converted into machine code by a translator (compiler or interpreter) before it can run (1) 2. Any two features, each named (1) + how it helps (1), e.g.: IntelliSense/auto-complete (1) – suggests keywords, variable and function names and shows parameters as you type, so fewer typing and syntax errors and faster coding (1) / syntax highlighting/colour coding (1) – keywords, strings and comments in different colours so code is easier to read and mistakes stand out (1) / syntax error highlighting (wavy underline) (1) – errors are spotted immediately as you type (1) / line numbering (1) – find a line quickly when an error message gives a line number (1) / code outlining (1) – collapse/expand blocks so long programs are easier to navigate (1) / auto-indent (1) – keeps indentation consistent so structure is clear (1) 3. The compiler translates the whole source program into machine code in one go before it runs (1) / producing a separate executable file; it reports all errors as a list after compiling and will not produce the executable until syntax errors are fixed (1)